

RESEARCH INTERESTS

I'm an M.S./Ph.D. student in Artificial Intelligence at KAIST AI, advised by Professor Seungryong Kim at [KAIST CVLAB](#). My research lies in computer vision, with a focus on World Models, Auto-Regressive Video Generation, and 4D Representations, aiming toward efficient and real-time world simulation.

My current research focuses on building world models that are grounded, persistent, embodied, and efficient. My recent work includes **real-world grounded world modeling** (e.g., **Seoul World Model**), **persistent world modeling** for maintaining world states beyond the current view (e.g., **World Observer**), **egocentric and multi-agent world modeling** for AR/VR (e.g., **Multi-Agent Egocentric World Model**), and **efficient autoregressive generation** for real-time simulation (e.g., **Speculative Forcing**).

WORK EXPERIENCE

NAVER AI Lab

Research Scientist Intern

Jul 2026 – Nov 2026

- Mentors: [Sangdoo Yun](#) and [Dongyoon Han](#).
- Developed efficient autoregressive video/world generation methods for real-time world simulation, leading to *Speculative Forcing for Efficient Autoregressive Video Generation* (ICLR 2027 Submission).

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Advisor: Seungryong Kim

M.S/Ph.D Integrated Student in Artificial Intelligence

Mar 2026-

Korea University

B.S in Cyber Defence (GPA: 4.33/4.5, **ranked 1st**)

Mar 2022-Feb 2026

- Full-funded Scholarship
- Vice President of Cyber Defence

RESEARCH EXPERIENCE

Undergraduate Research Intern

Advisor: Seungryong Kim

Kim Jaechul Graduate School of AI at KAIST

July 2024–Feb 2026

- Research in World Models, Video Generation, and Segmentation

PUBLICATIONS

World Observer: Joint Actor-Observer Generation for Persistent World Modeling

Hyunwook Choi, Dahyun Chung, Hyunsung Kim, Siyoon Jin, Jinhyeok Choi, Junyoung Seo, Seungryong Kim

ICLR 2027 Submission [\[Project Page\]](#)

Speculative Forcing for Efficient Autoregressive Video Generation

Hyunwook Choi, Seungryong Kim, Dongyoon Han, Sangdoo Yun

ICLR 2027 Submission [\[Project Page\]](#)

Multi-Agent Egocentric World Model with Fine-Grained Embodied Interaction

Dahyun Chung, Siyoon Jin, **Hyunwook Choi**, Honggyu An, Junyoung Seo, Hyunsung Kim, Seung Wook Kim, Seungryong Kim

ICLR 2027 Submission [\[Project Page\]](#)

Seoul World Model: Grounding World Simulation Models in a Real-World Metropolis

Junyoung Seo*, **Hyunwook Choi***, Minkyung Kwon, Jinhyeok Choi, Siyoon Jin, Gayoung Lee, Junho Kim, JoungBin Lee, Geonmo Gu, Dongyoon Han, Sangdoo Yun, Seungryong Kim, Jin-Hwa Kim

ECCV 2026 Oral; F2S@ICMLW 2026 Oral [[Project Page](#)]

MATRIX: Mask Track Alignment for Interaction-Aware Video Generation

Siyoon Jin, Seongchan Kim, Dahyun Chung, Jaeho Lee, **Hyunwook Choi**, Jisu Nam, Jiyoung Kim, Seungryong Kim

ICLR 2026 [[Project Page](#)]

Referring Video Object Segmentation via Language-aligned Track Selection

Seongchan Kim, Woojeong Jin, Sangbeom Lim, Heeji Yoon, **Hyunwook Choi**, Seungryong Kim

AdaptFM@ICMLW 2026 [[Project Page](#)]

S4M: Boosting Semi-Supervised Instance Segmentation with Segment Anything Model

Heeji Yoon, Heeseong Shin, Eunbeen Hong, **Hyunwook Choi**, Hansang Cho, Daun Jeong, Seungryong Kim

ICCV 2025 [[Project Page](#)]

ACADEMIC SERVICES

Reviewer

ICLR 2027

ICML 2026 Expo Talks and Panels

Speaker

July 2026

- Presented at the ICML 2026 Expo Talks and Panels. [[Talk Page](#)]

HONORS AND AWARDS

- National Science & Technology Scholarship (Ranked 1st in Department)
- Dean's List, Korea University (Spring 2024, GPA: 4.5/4.5)
- Semester High Honors, Korea University (Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024)
- 2nd Prize, Best Project, Korea University Deep Learning Association (Spring 2025)

TECHNICAL SKILLS

- Programming: Python, C++
- Deep Learning: Pytorch
- Synthetic Engine: Carla

INTERNATIONAL EXPERIENCE

London, United Kingdom

Oct 2010–Dec 2014